Chem 3A Lab Practical Grading Rubric

The Chem 3A lab practical assesses each student's ability to record data (qualitative and quantitative), and interpret data. Each station is worth an equal number of points (10). This rubric provides suggested point awards/deductions for each station when determining partial credit.

Version A - Green

Station	Example Ideal Response	Point Breakdown
1 **NaCl Standard Curve**	Mass Empty: 12.0772 g Mass Full: 22.2406 g Mass solution: 10.1634 g* Density: $\frac{10.1634\ g}{10.00\ mL} = 1.016\ g/mL$** *Student may or may not have determined mass of solution by difference	1 – Precision: Mass measured has 4 decimals 1 – Unit: Mass measured has "g" unit 4 – Accuracy: Mass of solution obtained is between +/- 0.1 g of expected/average 2 – Density is m/V 1 – Unit: Answer unit is g/mL or equivalent 1 – Sig Figs: Answer has 4 SF
2 **Synthesis of Alum**	$Percent\ Yield\ (\%) = \frac{7.4415\ g}{8.879\ g} \times 100\% = 83.81\ \%$ Yes, this percent yield is reasonable because it is under 100%. Masses are provided to students.	1 – Formula for % Yield 4 – Matched masses to Actual & Theoretical yield masses correctly 1 – Multiplied by 100 (dec to %) 1 – Sig Figs: Answer reported has 4 SF 1 – Unit: Wrote a % sign for answer 2 – Stated that the percent yield is reasonable
3 **Gas Production**	$0.0475\ g\ Mg \times \frac{1\ mol\ Mg}{24.34\ g} \times \frac{1\ mol\ H_2}{1\ mol\ Mg} \times \frac{2.016\ g\ H_2}{1\ mol\ H_2} = 0.00394\ g$	2 – Converted into mol Mg (1 if attempted but unsuccessful) 2 – Converted to mol H_2 (mol ratio) 2 – Converted to g H_2 (1 if attempted, but unsuccessful) 1 – Answer numerically correct 1 – Unit: Answer unit is g 1 – Sig Figs: Answer has 4 SF 1 – Setup as 3 conversions in a row
4 **Calorimetry**	23.1 ℃ ** Reasonable temperature, recorded to tenths place	3 – Accuracy: Reasonable value. 4 – Precision: Recorded to tenths place 3 – Unit: Measurement is ℃
5 **Making Measurements**	Measure volume based on dimensions Longest: 3.72 cm Short A: 1.14 cm Short B: 1.20 cm $V = 3.72\ cm \times 1.14\ cm \times 1.20\ cm = 5.09\ cm^3$	1 – Measurements have unit "cm" 2 – Recorded two short sides and one long side 2 – Precision: Recorded in hundredths place (no credit if measured in inches) 2 – Accuracy: Within 0.1 of key 1 – Multiplied to get V 1 – Answer has 3 SF 1 – Answer has unit cm^3

6 **Types of Reactions**	Observations: Milky white solid formed Reaction Type: Precipitation Equation: $CaCl_2(aq) + Na_2CO_3(aq) \rightarrow CaCO_3(s) + 2\,NaCl(aq)$	2 – Observations say white, milky, cloudy 2 – Precipitation (1 if DD) 1 – Wrote reactants from bottles 3 – Predicted Chemical formulas of products (-2 if one product correct) 1 – Balanced 1 – Correct phases
7 **Determination of Citric Acid**	Record buret reading, changes each semester Final Buret reading: **19.70 mL**** Innitial Buret reading: 1.27 mL Volume Added: $19.70\,mL - 1.27\,mL = 18.43\,mL$	1 – Recorded any volume from the buret 1 – Measurement has "mL" unit 2 – Accuracy: Recorded within 0.1 of true value 2 – Precision: Recorded to hundredths place 2 – Subtracted two volumes to get net added 1 – Answer has 2 decimal places 1 – Answer has mL unit
8 **General Lab Knowledge**	Bunsen Burner Graduated Cylinder	5 – Exact Name Correct (spelling excused) 3 - Wrong name, described purpose/use 2 – Partial name match, no description (i.e. Torch burner) 5 – Exact Name Correct (spelling excused) 3 - Wrong name, described purpose/use 2 – Partial name match, no description (i.e. Volumetric Cylinder)
9 **General Lab Knowledge**	$M_1V_1 = M_2V_2$ $M_2 = \frac{M_1V_1}{V_2} = \frac{(2.5789\,M)(25.00\,mL)}{100.00\,mL}$ $M_2 = 0.6447\,M$	4 – Strategy: Used $M_1V_1 = M_2V_2$ or molarity as conversion factor to get moles, then divided by volume to get new molarity. 2 – Matched V_1 and V_2 correctly 2 – Answer numerically correct 1 – Answer has unit "M" or "mol/L" 1 – Answer has 4 SF
10x **Empirical Formula**	Empirical Formula $22.1722\,g - 21.9751\,g = 0.1971\,g\,O$	2 - re-wrote data provided 2 - Identified mass of compound as "Weighing 2" (and didn't used Weighing 1) 2 – Subtracted two masses 2 – Numerically correct mass of O 1 – Answer has 4 decimal places 1 – Answer has "g" unit
10y **pH**	pH of Household items $[H_3O^+] = 10^{-pH}$ $[H_3O^+] = 10^{-2.5}$ $[H_3O^+] = 3.16 \times 10^{-3}\,M$	2 - Identified pH 4 - showed equation with numbers plugged in (2 if generic equation only) 1 – Numerically correct 1 – Answer has 1-3 SF 2 – Answer has units of M or mol/L

Version B – Yellow

Station	Example Ideal Response	Point Breakdown
1 **NaCl Standard Curve**	$\frac{51.9344\ g - 46.9067\ g}{50.00\ mL} \times 100$ = 10.06%	1 – Answer has 4 SF 1 – Reported answer with unit "%" 2 – Divide and multiply by 100 to get %m/v 3 – Identified Correct Volume 3 – Subtract to get mass of solute
2 **Synthesis of Alum**	$Full: 27.1868\ g$ $Empty: 17.2565\ g$ $Mass\ Alum: 9.9303\ g$ This measurement is the actual yield	2 – Recorded mass of full vial 1 – Recorded mass of empty vial 1 – Unit: Recorded mass(es) as "g" 2 – Subtracted two masses 2 – Sig Figs: Answer has 4 decimal places 2 – Noted that this is the "actual yield"
3 **Gas Production**	Volume of gas: $33.80\ mL$ ** (Read the eudiometer)	3 – Wrote down any volume of gas 3 – Precision: Recorded to hundredths place 2 – Unit: Recorded as "mL" 2 – Accuracy: within 0.2 mL of Instructor
4 **Calorimetry**	Temperature goes down Reaction was endothermic	4 – Observed temperature go down 6 – Conclusion of endo/exo consistent with observation
5 **Making Measurements**	Length: 3.84 cm $3.84\ cm \times \frac{10\ mm}{1\ cm} = 38.4\ mm$	2 – Wrote down any length 1 – Unit: recorded "cm" 2 – Accuracy: within 0.1 of key 2 – Precision: Recorded to hundredths place 1 – showed work or described conversion to mm 2 – Correct conversion to mm
6 **Types of Reactions**	Observations: Bubbles Reaction Type: Gas Evolution (or Acid/Base) Balanced Equation: $2HCl(aq) + Na_2CO_3(aq) \rightarrow 2\ NaCl(aq) + H_2O(l) + CO_2(g)$	2 – Observations say "bubbles" or "gas" 2 – Gas Evolution or Acid/Base (1 if DD) 1 – Wrote reactants from bottles 3 – Predicted Chemical formulas of products (-2 if one product correct) 1 – Balanced 1 – Correct phases
7 **Determination of Citric Acid**	$17.12\ mL - 1.27\ mL = 15.85\ mL$ $15.85\ mL \times \frac{0.9573\ mol\ NaOH}{1\ L} \times \frac{1\ mol}{3\ mol} \times \frac{1}{10.00\ mL} = 0.5058\ M$	2 – Found net volume of NaOH used 2 – Used molarity of NaOH to get mol NaOH 2 – Used molar ratio 1:3 2 – Divided by volume to get molarity 1 – Answer has 4 SF 1 – Answer has units of M or mol/L
8 **General Lab Knowledge**	Erlenmeyer flask Test Tube	5 – Exact Name Correct (spelling excused) 3 - Wrong name, described purpose/use 2 – Partial name match, no description (i.e. flask) 5 – Exact Name Correct (spelling excused) 3 - Wrong name, described purpose/use

		2 – Partial name match, no description (i.e. Test Cylinder Tube)
9 **General Lab Knowledge**	Volume: 27.4 mL ** (check graduated cylinder divisions) Temperature: 23.8 ℃ **	2 – Accuracy: Volume is within 0.3 of key 2 – Precision: Recorded to the correct precision for graduated cylinder in use. 1 – Unit: Reasonable value reported in mL 2 – Accuracy: Volume is within 0.3 of key 2 – Precision: Recorded to tenths place 1 – Unit: Reasonable value reported in ℃
10x **Empirical Formula**	$Mass\ \%\ O = \frac{0.1003\ g}{0.2527\ g} \times 100\% = 39.69\%$	1 – Identified Mass of O 1 – Identified Mass of compound 3 – Setup correct 2 – Converted decimal to percentage 2 – Answer reported to 4 SF 1 – Answer has % sign
10y **pH**	pH of Household items $pH + pOH = 14$ $2.5 + pOH = 14$ $pOH = 14 - 2.5 = 11.5$	2 - Identified pH 4 - showed equation with numbers plugged in (2 if generic equation only) 2 – Numerically correct 2 – Answer has 1 decimal place

Version C – Purple

Station	Example Ideal Response	Point Breakdown
1 NaCl Standard Curve	$5.0234\ g \times \frac{1\ mol}{58.44277\ g} \times \frac{1}{0.05000\ L}$ $= 1.719\ M$	3 – Converted from g to mol 2 – Converted mL to L 2 – Divided mol/L 1 – Numerically correct 1 – Answer has 4 SF 1 – Answer has mol/L or M unit
2 Synthesis of Alum	Mass of foil: 0.3294 g $0.3294\ g\ Al \times \frac{1\ mol\ Al}{26.98\ g\ Al} \times \frac{2\ mol\ Alum}{2\ mol\ Al} \times \frac{474.46\ g\ Alum}{1\ mol\ Alum}$ $= 5.793\ g\ Alum$	2 – Recorded mass of Al foil 1 – Unit: mass has "g" unit 2 – converted g Al to mol Al 2 – Converted mol Al to mol Alum (mol ratio) 2 – Converted mol Alum to g Alum 1 – Answer has 4 SF and "g" unit
3 Gas Production	Temperature: 22.8 °C ** Atmospheric Pressure: 764.3 mmHg** VP of H_2O: 21.068 mm Hg $P_{total} = P_{H_2} + P_{H_2O}$ $764.3\ mmHg - 21.068\ mm\ Hg$ $= 743.2\ mmHg$	1 – Precision: temperature to tenths place 1 – Accuracy: Reasonable value (± 0.3 °C) 1 – Unit: Temperature unit noted as °C 2- Recorded VP of H_2O for this temp recorded 2 – Recorded P_{total} 1 – Matched variables to equation correctly 1 – Numerically correct 1 – Answer has units of mmHg
4 Calorimetry	$q = mC\Delta T$ $q = 35.0392\ g \times 4.184\ \frac{J}{g\ °C} \times 21.1\ °C$ $q = 3090\ J\ or\ 3.09\ kJ$	2 – Used mass of solution in calculation 3 – Plugged in numbers correctly 3 – numerically correct 1 – Answer has 3 SF 1 – Answer has units of J or kJ
5 Making Measurements	Initial water volume: 39.1 mL Final water volume: 46.4 mL $46.4\ mL - 39.1\ mL = 7.3\ mL$	2 – Precision: Recorded volumes to tenths place 1 – Units: Recorded volumes with unit "mL" 2 – Subtracted two volumes 3 – Accuracy: Volume is between 6.9-7.7 mL 1 – Answer reported in mL 1 – Answer precise to same decimal as volumes
6 Types of Reactions	Observations: White solid formed Reaction Type: Precipitation Balanced Equation: $2Na_3PO_4(aq) + 3BaCl_2(aq) \rightarrow 2Ba_3(PO)_4(s) + 6\ NaCl\ (aq)$	2 – Observations say white, milky, cloudy 2 – Precipitation (1 if DD) 1 – Wrote reactants from bottles 3 – Predicted Chemical formulas of products (-2 if one product correct) 1 – Balanced 1 – Correct phases

7 **Determination of Citric Acid**	17.50 mL **	2 – Read any volume off Buret 4 – Accuracy: Recorded within 0.1 mL of key 3 – Precision: Recorded to hundredths place 1 – Unit: Answer reported in mL
8 **General Lab Knowledge**	Watch Glass Volumetric Pipet	5 – Exact Name Correct (spelling excused) 3 - Wrong name, described purpose/use 2 – Partial name match, no description (i.e. Clock glass) 5 – Exact Name Correct (spelling excused) 3 - Wrong name, described purpose/use 2 – Partial name match, no description (i.e. Pipet)
9 **General Lab Knowledge**	Length of line: 3.85 cm	4 – Accuracy: Within 0.05 of key 4 – Precision: Recorded to hundredths 2 – Unit: Unit reported as "cm"
10x **Empirical Formula**	$Mass\ \%\ Mg = \frac{0.1495\ g}{0.2479\ g} \times 100\% = 60.31\ \%$	1 – Identified Mass of Mg 1 – Identified Mass of compound 3 – Setup correct 2 – Converted decimal to percentage 2 – Answer reported to 4 SF 1 – Answer has % sign
10y **pH**	pH = 3.0	4 – Accuracy: Within ± 1 of key 4 – Precision: Recorded as x.0 or x.5 2 – No unit or other math